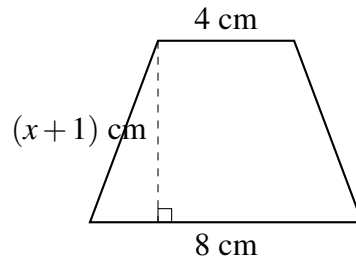
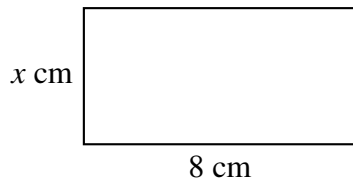


Area and Perimeter

- 1 A rectangle and a trapezium are shown below.



The area of the rectangle is equal to the area of the trapezium.

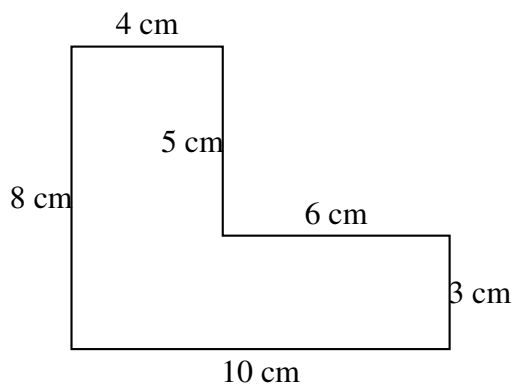
- (a) Find the value of x .

..... [3]

- (b) Calculate the perimeter of the rectangle.

..... [1]

- 2 The diagram shows a composite L-shaped polygon.
All internal angles are right angles.



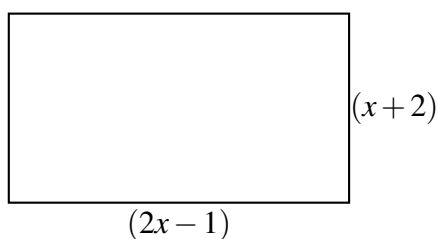
(a) Calculate the perimeter of the polygon.

..... [1]

(b) Calculate the total area of the polygon.

..... [3]

3 A rectangle is shown below. All dimensions are in centimetres.



The exact area of the rectangle is 18 cm^2 .

(a) Show that the area equation simplifies to:

$$2x^2 + 3x - 20 = 0$$

[2]

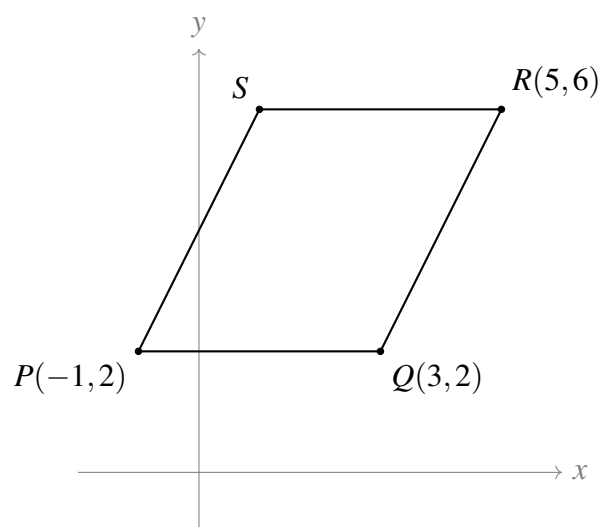
(b) Solve the equation to find the value of x .

..... [3]

- (c) Calculate the exact perimeter of the rectangle.

..... [2]

- 4 The points $P(-1,2)$, $Q(3,2)$, and $R(5,6)$ are three vertices of a parallelogram $PQRS$ drawn on a coordinate grid.



- (a) Write down the coordinates of the fourth vertex S .

..... [1]

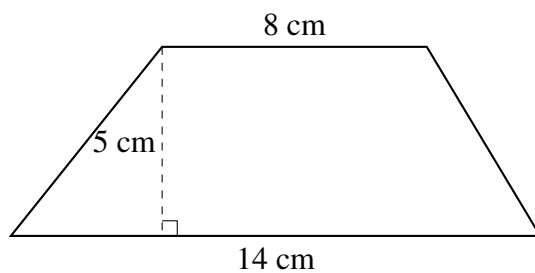
- (b) Calculate the area of the parallelogram $PQRS$.

..... [2]

- (c) Find the exact length of the diagonal PR . Give your answer in the form $a\sqrt{b}$ where a and b are prime numbers.

..... [3]

- 5 The diagram shows a trapezium.



- (a) Calculate the area of the trapezium.

..... [2]

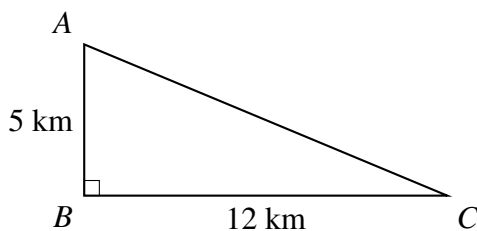
- (b) A distinct triangle has the same area as the trapezium. The triangle has a base of 11 cm and a perpendicular height of h cm.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Show that the height, h , of the triangle is 10 cm.

[2]

- 6 A sailing competition takes place around a triangular course ABC . The distance from mark A to mark B is 5 km, and the distance from B to C is 12 km. The angle at B is exactly 90° .



- (a) Calculate the total perimeter of the sailing course.

..... [3]

- (b) A leading boat completes the entire course at a constant average speed of 10 km/h.

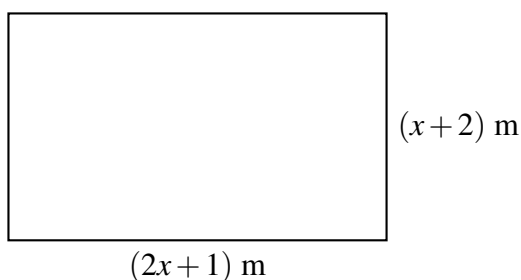
- (i) Calculate the time taken for the boat to finish, giving your answer in hours and minutes.

..... [2]

- (ii) Next year, the organizers plan to extend the course boundaries to form a mathematically similar triangle whose enclosed area is 44% larger than the current course. Calculate the perimeter of this new enlarged course.

..... [3]

- 7 A rectangular garden is shown in the diagram. The length of the garden is $(2x + 1)$ m and the width is $(x + 2)$ m. The total area of the garden is 27 m^2 .



(a) Show that $2x^2 + 5x - 25 = 0$.

[2]

(b) Solve the equation to find the exact value of x .

..... [2]

(c) Hence calculate the perimeter of the garden.

..... [2]

8 The parallel sides of a trapezium are in the ratio 3 : 5.



The perpendicular height of the trapezium is 8 cm.

The area of the trapezium is 96 cm^2 .

(a) Calculate the lengths of the two parallel sides.

..... [3]

(b) The trapezium is isosceles. Calculate the exact perimeter of the trapezium. Give your answer

in the form $p + q\sqrt{r}$ where p, q , and r are integers.

..... [4]